

CS 7646: Project 8 - Putting it All Together

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Abstract—Financial analysts have long been searching for ways to predict the future price of a stock. They have developed many "indicators" over time that are derived from historical data to help them make these predictions. Now, with the rise of machine learning, investors are more empowered than ever to make use of these indicators to create automated trading strategies. We will explore a Manual Strategy and a Machine Learning Strategy based off a Q-Learner to try and build profitable algorithms. We will compare the performance of these two strategies over historical data for JPM, and see how they perform in the presence of trading impact.

1 INTRODUCTION (OPTIONAL)

Trading "Indicators" are data points derived from historical data for a stock that can be used to predict future price movements. They are often used as inputs for trading strategies to determine when to buy or sell a stock. By using strategies that make decisions solely based on indicators, investors can automate their trading. While this practice can create a trading bot, it takes a lot of human time to develop a strategy that is profitable. However, with the advent of machine learning technologies, traders can now develop artificial intelligence algorithms to essentially design a strategy on its own. In this report, we hypothesize that a machine learning "strategy learner" will perform better than a human-designed strategy. To test this hypothesis, we will compare the performance of a human-designed strategy to a machine learning algorithm over historical pricing data for JPM in Experiment 1. Then, we will look at the effect trading "impact" has on our strategy learner. Trading "impact" is the subtle price difference that occurs when a trader buys or sells shares of a stock. A value of 0.005 is commonly chosen for calculations, and represents a 0.5% price difference. We hypothesize that a higher impact will cause our strategy portfolio to make less trades and result in a lower sharpe ratio for our portfolio. We will test this by running our strategy learner with varying impact values.

2 INDICATOR OVERVIEW

2.1 Percentage Price Oscillator (PPO)

Percentage Price Oscillator (PPO) is a momentum indicator that shows the relationship between the short-term and long-term moving averages of stock's price. To calculate it, we first subtract the long-term moving average from the short-term moving average. Then, we divide that value by the long-term moving average and multiply it by 100 to get a percentage.

We can tune the PPO by changing the short-term and long-term moving average periods. In general, positive PPO values indicate that the short-term moving average is above the long-term moving average, which may indicate positive momentum. For both the Manual Strategy and the Strategy Learner, we found that a short-term period of 5 days and a long-term period of 25 days worked well. We used the thresholds of -2.0 and 2.0 to make buy and sell decisions in our Manual Strategy.

2.2 Rate Of Change (ROC)

Rate of Change (ROC) is also a momentum indicator. This indicator shows how the price of a stock has changed over a certain period of time in order to identify trends. We calculate ROC for a stock by taking the current closing price and subtracting the closing price from n days ago. Then, we divide that value by the n-day-old price and multiply it by 100 to get a percentage. If ROC is positive, it means the stock price has increased over the period, which could indicate positive momentum. ROC may also indicate changes in trend directions when it crosses the zero line. For both the Manual Strategy and the Strategy Learner, we found that a period of 30 days worked well. We used the thresholds of -2.0 and 2.0 to make buy and sell decisions in our Manual Strategy.

2.3 Fast Stochastic Oscillator (%K)

The Fast Stochastic Oscillator, or %K, is a momentum indicator that compares a stock's closing price to its high and low prices for each day over some time. We calculate %K by first finding the highest and lowest prices seen over the last n days. Then, we subtract the lowest price seen from the current closing price, and divide that product by the difference between the highest and lowest prices. Finally, we multiply that value by 100 to get a percentage. The logic behind this indicator is

that when a stock is overbought, it will be trading near its high price, and when it is oversold, it will be trading near its low price. Therefore, a Fast Stochastic value of 80% or higher is often used to signal that a stock is overbought, while a value of 20% or lower is often used to signal that a stock is oversold. For both the Manual Strategy and the Strategy Learner, we found that a period of 30 days worked well. We used the thresholds of 20% and 80% to make buy and sell decisions in our Manual Strategy.

3 MANUAL STRATEGY

3.1 Description

We combined the indicators PPO, ROC, and %K to create a manual strategy. The logic of the strategy is to look for periods where both PPO and ROC agree on the trend of a stock price. Then, we look for the %K indicator to decide when to buy or sell, based on whether the stock is overbought or oversold. We believe by relying on all these signals to agree before making any orders, we can effectively make profitable trades.

In particular, we buy when:

- $PPO > -2.0$
- $ROC > -2.0$
- and $\%K < 20\%$

We sell when:

- $PPO < 2.0$
- $ROC < 2.0$
- and $\%K > 80\%$

3.2 Comparison and Evaluation

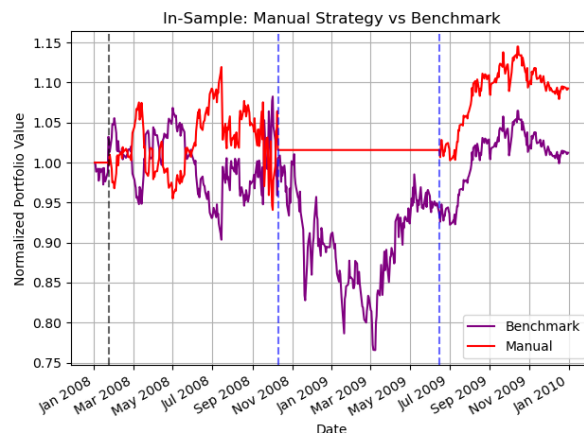


Figure 1—Manual Strategy vs Benchmark Strategy during in-sample period of 2008-209. Both portfolio values are normalized to 1. Blue vertical lines indicate LONG entrypoints, and Black lines indicate SHORT entrypoints.

Figure 1 shows the performance of our Manual Strategy when trading JPM during the in-sample period of Jan 1, 2008 to Dec 31, 2009. We compare this strategy to a Benchmark Portfolio, which buys 1000 JPM shares on the first trading day and holds. As we expected, our strategy outperforms the Benchmark Portfolio by a considerable margin. There were only about 2 months over the 24 month period where the Benchmark Portfolio was more valuable. One interesting thing to note is that the Manual Strategy value remained flat during the period where the Benchmark Strategy saw the largest dip. Over this period, the Manual Strategy remained profitable by only holding cash. This was a winning strategy during the in-sample period.

Of course, it is expected that our Manual Strategy performs better during the in-sample period as that is the data we used to tune the strategy. In particular, we tuned the lookback periods and thresholds until we found a strategy that is profitable for the in-sample period. However, we will next look at the performance of our strategy during the out-of-sample period of Jan 1, 2010 to Dec 31, 2011.

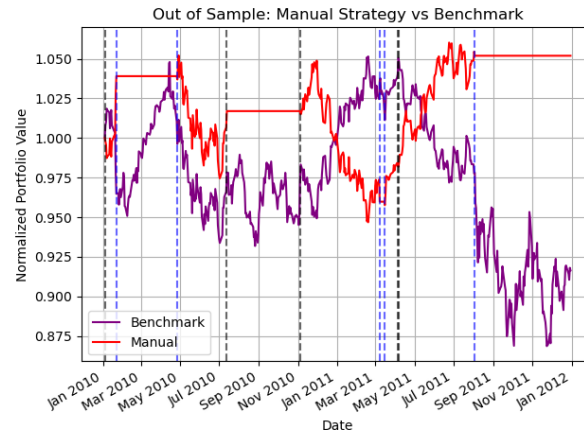


Figure 2—Manual Strategy vs Benchmark Strategy during out-of-sample period of 2010-2011. Both portfolio values are normalized to 1. Blue vertical lines indicate LONG entrypoints, and Black lines indicate SHORT entrypoints.

Figure 2 shows that once again, the Manual Strategy outperforms the Benchmark Portfolio. In fact, while the Benchmark Portfolio drops almost 10% by the out-of-sample period, the Manual Strategy manages to walk away with an overall profit above 5%. However, the Manual Strategy did spend more months with a value below that of the Benchmark Portfolio. In this case, it was about 6 months where the Benchmark Portfolio was worth more. Once again though, we see that the Manual Strategy only held cash at times when the Benchmark Portfolio dropped to it's lowest value.

Looking at these results, it may be easy to assume that the Manual Strategy is a good strategy, as the out-of-sample results outperformed even the in-sample-results. Again, this is not expected as we tuned the strategy to perform well on the in-sample data while being blind to out-of-sample data. Looking closely between both Figures 1 and 2 though, we see that the in-sample results look more consistent and stable than the out-of-sample results. It seems like the in-sample-results would be profitable whenever the time period ended, whereas the out-of-sample results seemed to get lucky by being profitable right at the end of the period. However, it is certainly promising that the Manual Strategy is also profitable during the out-of-sample period.

4 STRATEGY LEARNER

In order to see if we can build a better strategy than the Manual Strategy, we decided to use a Q-Learner to build our Strategy Learner. The Q-Learner is a reinforcement learning algorithm that takes a discrete state space as input. It emits 3 possible actions: buy, sell, or hold. During training, we iterate over the Q-Learner's state and suggested actions to see if they are profitable. We measure a value called "reward" based on how profitable the action was. This reward gets fed back into the Q-Learner to update the Q-values for every possible state that it sees.

In order to train our Q-Learner, we first needed to discretize our inputs, PPO, ROC, and %K. For PPO, we assign a discrete value 0-4 based on which bin the indicator falls into:

- < -2.0
- -2.0 to -0.5
- -0.5 to 0.5
- 0.5 to 2.0
- > 2.0

For ROC, we also assign a value 0-4 based on 5 bins:

- < -10.0
- -10.0 to -2.0
- -2.0 to 2.0
- 2.0 to 10.0
- > 10.0

Lastly, for %K, we assign a value 0-3 based on 4 bins:

- 0 to 25.0
- 25.0 to 50.0
- 50.0 to 75.0
- 75.0 to 100.0

After finding the 3 correct buckets for a given day, we calculate a discrete state using the formula:

$$\text{state} = (\text{roc}_{\text{bin}} * 20) + (\text{ppo}_{\text{bin}} * 4) + \text{percentk}_{\text{bin}}$$

Next, we developed a reward function for the Q-Learner. We calculated our reward by looking forward by 3 days to see if the action we took will be profitable or not. This profit calculation takes into account the commission and impact of the trade. By rewarding the Q-Learner when we notice profitable trades, we hope it will learn to continue that pattern in the future. Unprofitable trades similarly are penalized with a negative reward equal to the losses we project by looking forward 3 days. Lastly, we set the reward function to penalize the Q-Learner with a reward of -30 when it decides to HOLD. We found that the parameters already placed on our learners (that it can only hold positions -1000, 0, and 1000) made it difficult for the learner to make too many trades even if it sends many signals. With these boundaries in place, we figured it best to incentivize our learner to send as many BUY and SELL signals as possible.

With a discretized state space and a reward function, we were finally able to train our model over in-sample data. While training the Strategy Learner over the in-sample data, we manually tuned the parameters of the Q-Learner to find what produced the best results. We found the best results with a learning rate (alpha) of 0.15, a discount factor (gamma) of 0.85, a random action rate (rar) of 0.5, and a random action decay (rard) of 0.999. We did not find dyna hallucinations to be helpful.

We will compare the performance of our Q-Learner to the Manual Strategy in Experiment 1 below.

5 EXPERIMENT 1

5.1 Training Period

In Experiment 1, we compare our Manual Strategy to our Strategy Learner over the in-sample period of Jan 1, 2008 to Dec 31, 2009.

Figure 3 shows the performance of our Manual Strategy and the Strategy Learner alongside our Benchmark Portfolio which holds 1000 shares. As we hypothesized, the Strategy Learner outperforms the Manual Strategy by a significant margin with profit of 42% compared to 10%. The Manual Strategy beats the Benchmark Portfolio, which sees hardly any profit. While we are not surprised that the Strategy Learner beats the Manual Strategy which beats the Benchmark, it is interesting to see how well the Strategy Learner is able to perform. It is worth noting, however, that the Strategy Learner is at high risk of overfitting its model to the

in-sample-data that it trained on. Therefore, we should expect that the model does not perform as well when we look at out-of-sample data.

5.2 Testing Period

To see if the Strategy Learner can actually be profitable, we must test it against out-of-sample data as well.

Figure 4 shows that once again the Strategy Learner performs best, followed by the Manual Strategy, and finally the Benchmark Portfolio. In this case, the Strategy Learner only makes a profit of 10% over the 2 year period, while the Manual Strategy makes a profit of 5%. Both of these strategies are winning for the investor, and especially beat out the Benchmark Portfolio which ends with an 8% loss.

In general, we'd expect that the Strategy Learner would perform best over most periods of data. The Strategy Learner is able to be more adaptable to our indicators than the Manual Strategy, and can learn over thousands of quick iterations instead of having an operator manually tuning every parameter. However, with the unpredictable nature of the stock market, we are also sure that we can find time periods where our strategies don't perform as well. In order to ensure a strategy can remain positive in the future, it would be valuable to test across many different time periods (this is an opportunity for future follow-up).

6 EXPERIMENT 2

TODO

Handgloves
Handgloves

Figure 3—Palatino. Make sure the live text (bottom) uses the same font as the image (top).

6.1 Body text

Body text is set in the regular weight at 11 points with custom line spacing of 1.26, and 8.5 points of spacing added after each paragraph. It should be justified and paragraphs should not be indented. These styles can be automatically applied using the *Normal* paragraph style.

Bold and *italics* should be used for emphasis. Hyperlinks may be inserted in the text, as well as in-line code, superscripts^{like this}, and subscripts_{like this}.

6.2 Title & subtitle

The paper title should be set in the regular weight at 17 points with 22 points of line spacing, centered at the top of the first page. The title may span up to three lines. For typical assignments, the document title may be as simple as "Assignment 1" More specialized assignments may warrant more unique paper names, like "A Proposal to Create a New Document Format."

The author's name and email should come next unless you want to or were asked to submit anonymously, in which case this can be omitted. They should be set in the same size and weight as body text, centered. These styles can be applied using the *Title* and *Subtitle* paragraph styles.

6.3 Abstract

If your paper requires an abstract, it should be placed at the top of the first page underneath the title block, preceded by the word *Abstract* in bold italic. An extra 0.5" should be added to both sides. Not all papers require abstracts; only those that would benefit from a high-level summary of the project or its background.

6.4 Headings

Headings should all be set in the same size as body text (11 pt) in bold. With the exception of *Heading 1*, they should all have 8.5 pt of space before and after. They should be hierarchically numbered: Word, Pages, and L^AT_EX will do this automatically when you use the appropriate paragraph styles, but Google Docs users will need to number their headings manually. Headings should not span more than one line.

6.4.1 *Heading 1*

Heading 1 should be set in all caps. It should have 11 points of space added before and 8.5 points of space added after

6.4.2 *Headings 2–4*

Besides *Heading 1*, which is set in caps, headings should always use sentence case (i.e., first word capitalized) rather than title case; after all, they are not titles.

Heading 2 should be set in bold roman (upright), and *Heading 3* should be set in bold italics. The use of headings beyond *Heading 3* is discouraged.

Heading 4—*Heading 4* is provided as a run-in sidehead. Like *Heading 3*, it is set in bold italics, but it should be followed by an em dash and flow right into the text, as seen at the beginning of the current paragraph. It should be used more as a list style than a heading style, e.g. to set off a list of principles in a heuristic evaluation.

6.5 Page layout

JDF uses the US Letter paper size (8.5" x 11"). It has a top margin of 1", and bottom and side margins of 1.5". This yields a text block of 5.5" x 8.5", which is exactly $\frac{1}{2}$ the size of the page, divided lengthwise.

The page number should be included in the bottom margin, 1" from the bottom of the page. No other elements should be placed in the margins.

7 PRESENTATIONAL ELEMENTS

You are encouraged to use presentational elements liberally, as long as they add to the clarity of your submissions. They often require less space and fewer accompanying words to explain a given concept, and do a far better job of it.

7.1 Figures

Figures should always be centered on the page, although they may also take up the entire width and height of the text block. Figures should always be referenced in the text, and they should include a descriptive caption. Figures may also be equations, diagrams, or other kinds of content.

If your figure includes a white background (e.g. an interface design or graph), it may aid legibility to add a $\frac{1}{4}$ -point black border.

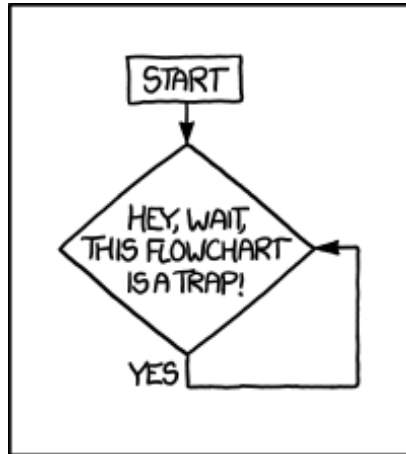


Figure 4—Make sure your flowcharts are more useful than this one. Source: [XKCD](#).

Figure captions should be placed beneath the corresponding figure, indented 1" on the left and right sides. The label for the figure, e.g. "Figure 1," should be set in bold italics followed by an em dash, and the entire caption should be 8.5 points with 14 points of line spacing. Again, Word and LaTeX will number these automatically using the *Figure Caption* paragraph style, but Docs users will need to number these manually. If need be, you may have one figure caption corresponding to multiple consecutive figures and use either locational descriptors (e.g. "top left," "middle") or labels (e.g. "A", "B") to map parts of the caption to parts of the figure. Make sure that caption falls on the same page as the corresponding figure or table; you may rearrange text to make this work.

In Microsoft Word, you may need to either change the image's text wrap settings to "Top and Bottom" or change the line spacing of the image to 1.0.

7.2 Tables

You have freedom to format tables in the way that works best for your data. Generally, text should be left-aligned and numbers should be right-aligned or aligned at the decimal – you can do this using a [custom tab stop](#). The default table style (shown below) reduces the text size to be equal to the caption text.

Table captions should be formatted the same way as figure captions, but they should be placed above the table. The popular mnemonic for this is: figures at the foot, tables at the top. Like figures, tables should not exceed the margins and should be centered on the page.

Table 1—Mathematical constants. Notice how the approximations align at the decimal.

Name	Symbol	Approximation	Description
Golden ratio	ϕ	1.618	Number such that the ratio of " to the number is equal to the ratio of its reciprocal to 1
Euler's number	e	2.71828	Exponential growth constant
Archimedes' constant	π	3.14	The ratio between circumference and diameter of a circle
One hundred	A+	100.00	The grade we hope you'll all earn in this class

7.3 Additional elements

There are additional elements you may want to include in your paper, such as in-line or block quotes, lists, and more. For other content types not covered here, you have flexibility in determining how it should be used in this format.

7.3.1 Quotes

If you would like to quote an outside source, you may do so with quotation marks followed by a citation. If a quote is fewer than three lines, you may write it in-line. It is acceptable to replace pronouns with their target in brackets for clarity. For example, "Heavy use of peer grading would compromise [the school's] reputation" (Joyner et al., 2016). If a quote exceeds three lines, you should set it as its own paragraph with 0.5" side margins, using the *Blockquote* style.

"Whether or not the grades generated by peers are reliably similar to grades generated by experts is only one factor worth considering, however. Student perception is also an important factor. A recent study indicated that reliance on peer grading is one of the top drivers of high MOOC dropout rates. This problem may be addressed by reintroducing some expert grading where possible." (Joyner et al., 2016)

7.3.2 Lists

Bulleted and numbered lists are indented 0.5" from the left margin, with the bullet or number hanging in the margin by 0.25" (the default format).

Bullet points:

- First bullet point item

- Second bullet point item

Numbered list:

1. First numbered item
2. Second numbered item

8 PROCEDURAL ELEMENTS

8.1 In-line citations

Articles or sources to which you refer should be cited in-line with the authors' names and the year of publication.¹ The citation should be placed close in the text to the actual claim, not merely at the end of the paragraph. For example: students in the OMSCS program are older and more likely to be employed than students in the on-campus program (Joyner, 2017). In the event of multiple authors, list them. For example: research finds sentiment analysis of the text of OMSCS reviews corresponds to student-assigned ratings of the course (Newman and Joyner, 2018). You may also cite multiple studies together. For example: several studies have found students in the online version of an undergraduate CS1 class performed equally with students in a traditional version (Joyner, 2018a; Joyner, 2018b). If you would like to refer to an author in text, you may also do so by including the year (in parentheses) after the author's name in the text. If a publication has more than 4 authors, you may list the first author followed by 'et al.' For example: Joyner et al. (2016) claim that a round of peer review prior to grading may improve graders' efficiency and the quality of feedback given. This applies to parenthetical citations as well, e.g. (Joyner et al., 2016).

8.2 Reference lists

References should be placed at the end of the paper in a dedicated section. Reference lists should be numbered and organized alphabetically by first author's last name. If multiple papers have the same author(s) and year, you may append a letter to the end of the year to allow differentiated in-line text (e.g. Joyner, 2018a and Joyner, 2018b in the section above). If multiple papers have the same author(s), list them in chronological order starting with the older paper. Only works that are

¹ In-line citations are preferred over footnotes, and we favor APA citation format for both in-line citations and reference lists. Refer to the [Purdue Online Writing Lab](#), or follow the above examples. Footnotes should use 8.5 point text with 1.26 line spacing.

cited in-line should be included in the reference list. The reference list does not count against the length requirements.

9 REFERENCES

1. Joyner, D. A., Ashby, W., Irish, L., Lam, Y., Langston, J., Lupiani, I., Lustig, M., Pettoruto, P., Sheahen, D., Smiley, A., Bruckman, A., and Goel, A. (2016). "Graders as Meta-Reviewers: Simultaneously Scaling and Improving Expert Evaluation for Large Online Classrooms". In: *Proceedings of the Third Annual ACM Conference on Learning at Scale*. Edinburgh, Scotland.
2. Joyner, D. A. (2017). "Scaling Expert Feedback: Two Case Studies". In: *Proceedings of the Fourth Annual ACM Conference on Learning at Scale*. Cambridge, Massachusetts.
3. Joyner, D. A. (2018a). "Intelligent Evaluation and Feedback in Support of a Credit-Bearing MOOC". In: *Proceedings of the 19th International Conference on Artificial Intelligence in Education*. London, United Kingdom: Springer.
4. Joyner, D. A. (2018b). "Toward CS1 at Scale: Building and Testing a MOOC-for-Credit Candidate". In: *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*. London, United Kingdom: ACM Press.
5. Newman, H. and Joyner, D. A. (2018). "Sentiment Analysis of Student Evaluations of Teaching". In: *Proceedings of the 19th International Conference on Artificial Intelligence in Education*. London, United Kingdom: Springer.

10 APPENDICES

You may optionally move certain information to appendices at the end of your paper, after the reference list. If you have multiple appendices, you should create a section with a *Heading 1* of "Appendices." Each appendix should begin with a descriptive *Heading 2*; appendices can thus be referenced in the body text using their heading number and description, e.g. "Appendix 5.1: Survey responses." If you have only one appendix, you can label it with the word "Appendix" followed by a descriptive title, e.g., "Appendix: Survey responses."

These appendices do not count against the page limit, but they should not contain any information required to answer the question in full. The body text should be sufficient to answer the question, and the appendices should be included only for you to reference or to give additional context. If you decide to move content to an

appendix, be sure to summarize the content and note it in relevant place in the body text, e.g., "The raw data can be viewed in *Appendix 5.1: Survey responses*."